Basics of identification theory

Lecture 3. Stochastic identification

Alexey Vedyakov

ITMO University

Table of contents

1. Least-squares method. Stochastic problem

2. Subtotals

3. Computational aspects

Least-squares method. Stochastic problem

Some concepts from probability theory

Let $v \in \mathbb{R}^1$ be some stationary random variable.

- Probability density $\rho(v)$: $\int_{-\infty}^{\infty} \rho(v)dv = 1$
- Expected value $\mathbb{E}\{v\} = \int_{-\infty}^{\infty} v\rho(v)dv = 1$. For N measurement $E_v(N) = \frac{1}{N}\sum_{k=1}^{N} v(k), E_v(N) \to \mathbb{E}\{v\}$ with $N \to \infty$
- Variance $\mathbb{D}\{v\} = \mathbb{E}\{(v - \mathbb{E}\{v\})^2\} = \sigma_v^2$. if $v \in \mathbb{R}^m$, then $\mathbb{D}\{v\} \in \mathbb{R}^{m \times m}$ - variance matrix.

Some concepts from probability theory

Let $v \in \mathbb{R}^1$, $\mu \in \mathbb{R}^1$ be some stationary random variables.

- Covariance: $\text{cov}(v, \mu) = \mathbb{E}\{(v - \mathbb{E}\{v\})(\mu - \mathbb{E}\{\mu\})\}$.
  $\text{cov}(v, v) = \mathbb{D}\{v\}$. For two vectors, the covariance matrix is obtained.
- Correlation: $r_{v,\mu} = \text{corr}(v, \mu) = \dfrac{\text{cov}(v, \mu)}{\sigma_v \sigma_\mu}$.
  $\text{corr}(v, v) = 1$
- Autocorrelation function:
  $$R_v(\tau) = \frac{\mathbb{E}\{(v(k) - \mathbb{E}\{v\})(v(k+\tau) - \mathbb{E}\{v\})\}}{\sigma_v^2}$$
  $$R_v(\tau) = R_v(-\tau), \qquad R_v(0) = 1.$$

Some concepts from probability theory

If variables v and μ are interdependent if $\rho(v|\mu) = \rho(v)$, then:

- $\mathbb{E}\{v, \mu\} = \mathbb{E}\{v\}\mathbb{E}\{\mu\}$
- $\text{cov}(v, \mu) = 0$
- $\text{corr}(v, \mu) = 0$

If variables v and μ are independent for $k_1 \neq k_2$ (not an autocorrelated process), then

- $\text{cov}(V, V) = \mathbb{D}\{V\} = \sigma_v^2 \mathcal{I}$
- $R_v(\tau) = 0$ for $\tau \neq 0$

Problem statement

Problem

Suppose that there is a vector of true values of $\boldsymbol{\theta}_0$ and that the measurement results are subject to noise:

$$y(k) = \mathbf{x}^T(k)\boldsymbol{\theta}_0 + v(k),$$

where v is random variable.

Necessary

A set of available measurements Y, X must be used to generate an estimate $\hat{\boldsymbol{\theta}}$ such that $\mathbb{E}\{\hat{\boldsymbol{\theta}}\} = \boldsymbol{\theta}_0$. It is desirable that the variance of $\hat{\boldsymbol{\theta}}$ is minimal.

Problem statement

Problem

Given: $y(k) = \mathbf{x}^T(k)\boldsymbol{\theta}_0 + v(k)$
Goal: $\hat{\boldsymbol{\theta}}$ such that $\mathbb{E}\{\hat{\boldsymbol{\theta}}\} = \boldsymbol{\theta}_0$

Hypotheses

Required

H1 $\mathbb{E}\{v\} = 0$

H2 v is interdependent with all x_i. For example, the regressor $\mathbf{x}$ is deterministic.

H3 $\det(X^T X) \neq 0$

Useful

H4 $\mathbb{D}\{v\} = \sigma_v^2 = \text{const}$

H5 $v(k)$ is not an autocorrelated process, i.e. separate $v(i)$ and $v(j)$ are interdependent for $i \neq j$, $\mathbb{D}\{V\} = \sigma_v^2 \mathcal{I}$

LSM, Theorem 2

Solution of a stochastic problem

Let a linear estimate be formed as $\hat{\boldsymbol{\theta}}_{LS} = (X^T X)^{-1} X^T Y$. Then if

- $\mathbb{E}\{v\} = 0$
- v and $\mathbf{x}$ are interdependent and performing $\mathbb{E}\{vx_i\} = \mathbb{E}\{v\}\mathbb{E}\{x_i\}$,

the estimate $\hat{\boldsymbol{\theta}}_{LS}$ is unbiased, i.e. $\mathbb{E}\{\hat{\boldsymbol{\theta}}_{LS}\} = \boldsymbol{\theta}_0$.
Moreover, if

- $\mathbb{D}\{v\} = \sigma_v^2 = \text{const}$
- $\mathbb{D}\{V\} = \sigma_v^2 \mathcal{I}$, i.e. $R_v(\tau) = 0$ for $\tau \neq 0$.

then $\hat{\boldsymbol{\theta}}_{LS}$ is an effective estimate, i.e. best in linear class
(BLUE - Best Linear Unbiased Estimation)

LSM, Theorem 2

Solution of a stochastic problem

Let a linear estimate be formed as $\hat{\boldsymbol{\theta}}_{LS} = (X^T X)^{-1} X^T Y$. Then if

- $\mathbb{E}\{v\} = 0$
- v and $\mathbf{x}$ are interdependent and performing $\mathbb{E}\{vx_i\} = \mathbb{E}\{v\}\mathbb{E}\{x_i\}$,

the estimate $\hat{\boldsymbol{\theta}}_{LS}$ is unbiased, i.e. $\mathbb{E}\{\hat{\boldsymbol{\theta}}_{LS}\} = \boldsymbol{\theta}_0$.
Moreover, if

- $\mathbb{D}\{v\} = \sigma_v^2 = \text{const}$
- $\mathbb{D}\{V\} = \sigma_v^2 \mathcal{I}$, i.e. $R_v(\tau) = 0$ for $\tau \neq 0$.

then $\hat{\boldsymbol{\theta}}_{LS}$ is an effective estimate, i.e. best in linear class
(BLUE - Best Linear Unbiased Estimation)

Linearity

$$Y = Y_1 + Y_2 \Rightarrow \hat{\boldsymbol{\theta}} = \hat{\boldsymbol{\theta}}_1 + \hat{\boldsymbol{\theta}}_2 \quad \text{and} \quad Y = \alpha Y_1 \Rightarrow \hat{\boldsymbol{\theta}} = \alpha \hat{\boldsymbol{\theta}}_1$$

Proof

Let us show the unbiased of the estimate. Indeed

$$\begin{aligned}\mathbb{E}\{\hat{\boldsymbol{\theta}}_{LS}\} &= \mathbb{E}\{(X^T X)^{-1} X^T Y\} = \mathbb{E}\{(X^T X)^{-1} X^T (X\boldsymbol{\theta}_0 + V)\} \\ &= \boldsymbol{\theta}_0 + (X^T X)^{-1} \mathbb{E}\{X^T V\}.\end{aligned}$$

Since **x** and v are interdependent and $\mathbb{E}\{v\} = 0$, then $\mathbb{E}\{X^T V\} = 0$ and $\mathbb{E}\{\hat{\boldsymbol{\theta}}_{LS}\} = \boldsymbol{\theta}_0$.

Proof

Let us show the unbiased of the estimate. Indeed

$$\begin{aligned}\mathbb{E}\{\hat{\boldsymbol{\theta}}_{LS}\} &= \mathbb{E}\{(X^T X)^{-1} X^T Y\} = \mathbb{E}\{(X^T X)^{-1} X^T (X\boldsymbol{\theta}_0 + V)\} \\ &= \boldsymbol{\theta}_0 + (X^T X)^{-1} \mathbb{E}\{X^T V\}.\end{aligned}$$

Since **x** and v are interdependent and $\mathbb{E}\{v\} = 0$, then $\mathbb{E}\{X^T V\} = 0$ and $\mathbb{E}\{\hat{\boldsymbol{\theta}}_{LS}\} = \boldsymbol{\theta}_0$. Consider the variance of the estimate

$$\begin{aligned}\mathbb{D}\{\hat{\boldsymbol{\theta}}_{LS}\} &= \mathbb{E}\{(\hat{\boldsymbol{\theta}}_{LS} - \boldsymbol{\theta}_0)(\hat{\boldsymbol{\theta}}_{LS} - \boldsymbol{\theta}_0)^T\} = \mathbb{E}\{(X^T X)^{-1} X^T V V^T X (X^T X)^{-1}\} \\ &= (X^T X)^{-1} X^T \mathbb{E}\{V V^T\} X (X^T X)^{-1} = \boxed{\sigma_v^2 (X^T X)^{-1}}.\end{aligned}$$

Proof

Let us now show that this variance is minimal (in the sense of a quadratic form) among all unbiased linear estimates. Consider some unbiased linear estimate $\boldsymbol{\theta} = MY$ with some M matrix.

Proof

Let us now show that this variance is minimal (in the sense of a quadratic form) among all unbiased linear estimates. Consider some unbiased linear estimate $\boldsymbol{\theta} = MY$ with some M matrix.
Since $\mathbb{E}\{\boldsymbol{\theta}\} = \boldsymbol{\theta}_0$, then $\mathbb{E}\{MX\boldsymbol{\theta}_0\} + \mathbb{E}\{MV\} = \boldsymbol{\theta}_0$ and $MX = \mathcal{I}$. With this $\mathbb{D}\{\boldsymbol{\theta}\} = \mathbb{E}\{MVV^T M^T\} = \sigma_v^2 MM^T$.

Proof

Let us now show that this variance is minimal (in the sense of a quadratic form) among all unbiased linear estimates. Consider some unbiased linear estimate $\boldsymbol{\theta} = MY$ with some M matrix.
Since $\mathbb{E}\{\boldsymbol{\theta}\} = \boldsymbol{\theta}_0$, then $\mathbb{E}\{MX\boldsymbol{\theta}_0\} + \mathbb{E}\{MV\} = \boldsymbol{\theta}_0$ and $MX = \mathcal{I}$. With this $\mathbb{D}\{\boldsymbol{\theta}\} = \mathbb{E}\{MVV^T M^T\} = \sigma_v^2 MM^T$.
Consider the difference $Q := \mathbb{D}\{\boldsymbol{\theta}\} - \mathbb{D}\{\hat{\boldsymbol{\theta}}_{LS}\}$ and show that $Q \geq 0$.

$$
\begin{aligned}
Q &= \sigma_v^2 \left(MM^T - \mathcal{I}(X^T X)^{-1}\mathcal{I}^T\right) = \sigma_v^2 \left(MM^T - MX(X^T X)^{-1}X^T M^T\right) \\
&= \sigma_v^2 M(\mathcal{I} - P)M^T,
\end{aligned}
$$

where $P := X(X^T X)^{-1}X^T$, and $PP^T = P = P^T$.

Proof

Moreover, $(\mathcal{I} - P)(\mathcal{I} - P)^T = \mathcal{I} - P$. Then

$$Q = \sigma_v^2 M(\mathcal{I} - P)(\mathcal{I} - P)^T M^T = \sigma_v^2 S S^T,$$

where $S \in \mathbb{R}^{m \times N}$, $S := M(\mathcal{I} - P)$.

Proof

Moreover, $(\mathcal{I} - P)(\mathcal{I} - P)^T = \mathcal{I} - P$. Then

$$Q = \sigma_v^2 M(\mathcal{I} - P)(\mathcal{I} - P)^T M^T = \sigma_v^2 S S^T,$$

where $S \in \mathbb{R}^{m \times N}$, $S := M(\mathcal{I} - P)$.
Then $Q \geq 0$ and

$$\boxed{\mathbb{D}\{\hat{\boldsymbol{\theta}}_{LS}\} \leq \mathbb{D}\{\boldsymbol{\theta}\}}$$

Subtotals

Least-squares method. Deterministic problem

For a given X, Y it is required to find the best approximation in the class of linear models $Y = X\theta$, in the sense of the least-squares criterion

$$V(\theta) = (Y - x\theta)^T (Y - x\theta).$$

Hypothesis: $\det(X^T X) \neq 0$.
Solution:

$$\boxed{\hat{\theta}_{LS} = (X^T X)^{-1} X^T Y}$$

minimizes the specified criterion, while

$$V(\hat{\theta}_{LS}) = Y^T Y - Y^T X (X^T X)^{-1} X^T Y$$

Least-squares method. Stochastic problem

For model

$$y(k) = \mathbf{x}^T(k)\boldsymbol{\theta}_0 + v(k)$$

an unbiased estimate of the $\boldsymbol{\theta}_0$ parameter vector should be generated.
Hypothesis: $\mathbb{E}\{v\} = 0$, $(\mathbf{x}^T(k), v(k))$ do not correlate, $\mathbb{D}\{V\} = \sigma_v^2\mathcal{I}$.
Solution:

$$\boxed{\hat{\boldsymbol{\theta}}_{LS} = (X^T X)^{-1} X^T Y}$$

is an unbiased estimate, i.e. $\mathbb{E}\{\hat{\boldsymbol{\theta}}_{LS}\} = \boldsymbol{\theta}_0$. If additionally $\mathbb{D}\{V\} = \sigma_v^2\mathcal{I}$, then

$$\boxed{\mathbb{D}\{\hat{\boldsymbol{\theta}}_{LS}\} = \sigma_v^2 (X^T X)^{-1}}$$

and estimation $\hat{\boldsymbol{\theta}}_{LS}$ is effective

Computational aspects

Computational aspects

The estimate precision also can be affected by calculation errors or more precisely rounding errors.

Consider the matrix regression

$$Y = X\theta, \tag{1}$$

where $Y \in \mathbb{R}^m$, $X \in \mathbb{R}^{m \times n}$, $\theta \in \mathbb{R}^n$.

The direct approach is finding estimate via solving normal equations

$$X^\top X\theta = X^\top Y. \tag{2}$$

However, this approach is sensitive to numerical rounding errors

Computational aspects. Example

Numerical rounding errors and solution of a system of linear equations.
Consider the following system

$$\begin{bmatrix} 3 & 3-\delta \\ 4 & 4+\delta \end{bmatrix} \theta = \begin{bmatrix} -1 \\ 1 \end{bmatrix}. \tag{3}$$

The exact solution is

$$\begin{bmatrix} -\frac{1}{\delta} \\ \frac{1}{\delta} \end{bmatrix}. \tag{4}$$

The normal equations $X^\top X\theta = X^\top Y$

$$\begin{bmatrix} 25 & 25+\delta \\ 25+\delta & 25+2\delta+2\delta^2 \end{bmatrix} x = \begin{bmatrix} 1 \\ 1+2\delta \end{bmatrix}. \tag{5}$$

Computational aspects. Example

The following system of

$$\begin{bmatrix} 25 & 25+\delta \\ 25+\delta & 25+2\delta+2\delta^2 \end{bmatrix} x = \begin{bmatrix} 1 \\ 1+2\delta \end{bmatrix}. \tag{6}$$